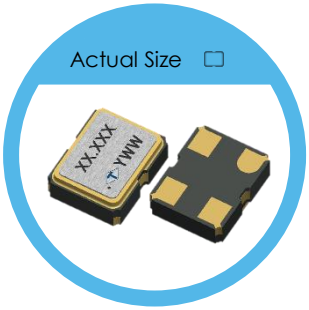
TAITIEN
Our Performance, Your Reputation

PY-U Type Frequency up to 200MHz

FASTXO 2.5 x 2.0 mm SMD Crystal Oscillator



RoHS Compliant

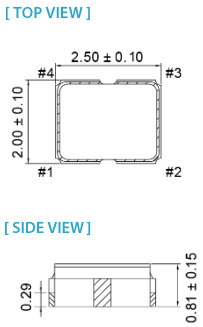
FEATURES

- Conforms to AEC-Q100
- Typical 2.5 x 2.0 x 0.81mm Ceramic SMD Package
- Operation Supply Voltage: 1.8V, 2.5V, 3.3V
- FASTXO Series, Fast Delivery at Any Frequency
- Frequency Stability ±20ppm over -40˚C to 105˚C
- Tri-State Enable/Disable
- Pb-free/RoHS Compliant

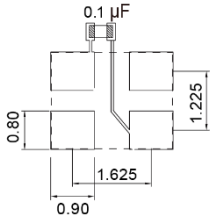
TYPICAL APPLICATION

- Computer Peripherals
- Set-top Box, HDTV, DSC, PDA

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between V_{DD} and GND pads.

PIN FUNCTION (mm)

PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

ELECTRICAL SPECIFICATION

Parameter		3.3V		2.5V		1.8V		Unit	Test Condition
		Min.	Max.	Min.	Max.	Min.	Max.		
Supply Voltage Variation (V_{DD})		V_{DD} – 5%	V_{DD} + 5%	V_{DD} – 5%	V_{DD} + 5%	V_{DD} – 5%	V_{DD} + 5%	V	
Frequency Range		1	200	1	200	1	125	MHz	
V_{DD} Sensitivity (±5%)		-2	2	-2	2	-2	2	ppm	
Supply Current (@15pf Loading)	1 MHz ≦ Fo < 30 MHz		27		27		25	mA	
	30 MHz ≦ Fo < 75 MHz		27		27		25	mA	
	75 MHz ≦ Fo < 125 MHz		30		30		25	mA	
	125 MHz ≦ Fo < 170 MHz		35		35			mA	
	170 MHz ≦ Fo ≦ 200 MHz		40		35			mA	
Output Level	Output High	90% V_{DD}		90% V_{DD}		90% V_{DD}		V	
	Output Low		10% V_{DD}		10% V_{DD}		10% V_{DD}	V	
Transition Time: Rise/Fall Time	1 MHz ≦ Fo < 10 MHz		3		4		4	nSec	
	10 MHz ≦ Fo < 125 MHz		2		3		3	nSec	
	125 MHz ≦ Fo < 200 MHz		2		2			nSec	
Duty Cycle		45	55	45	55	45	55	%	
Startup Time			8		8		8	mSec	
Tri-State	Output Enable	0.7 x V_{DD}		0.7 x V_{DD}		0.7 x V_{DD}		V	
	Output Disable		0.3 x V_{DD}		0.3 x V_{DD}		0.3 x V_{DD}	V	
RMS Phase Jitter (offset 12 kHz~20 MHz @ 50MHz)	Integer Mode		1.5		1.5		1.5	pSec	
	Fractional Mode		2.0		2.0		2.0	pSec	
Stand by Current (@PD Mode)			400		400		400	µA	
Stand by Current (@OE Mode)			25		25		25	mA	
Output Loading		15		15		15		pf	
Aging (@25˚C, 1st year)			±3		±3		±3	ppm	
Storage Temp. Range		-55	+125	-55	+125	-55	+125	˚C	

FREQ. STABILITY vs. TEMP. RANGE

ppm / Temp.(°C)	±15	±20	±25	±50
-20 ~ +70	△	○	○	○
-40 ~ +85	×	△	○	○
-40 ~ +105	×	△	○	○
-40 ~ +125	×	×	×	○

○: Available △:Conditional ×: Not Available
Inclusive of calibration @ 25˚C ,operating temperature rrange,input Voltage variation,load variation,aging (1st year),shock,and vibration

Note: not all combination of options are available. Other specifications may be available upon request.
Specifications subject to change without notice.

www.taitien.com
sales@taitien.com.tw